SECTION 08360 [08 36 00]
SECTIONAL OVERHEAD DOORS
THERMACORE® MODEL 591 INSULATED SECTIONAL STEEL DOORS

Display hidden notes to specifier by using 'Tools'/'Options'/'View'/'Hidden Text'. On newer versions of Microsoft Word click on round Windows logo in top left corner, Click on 'Word Options' button at bottom of drop down menu. Click on 'Display' on left menu bar, and check the box for 'Hidden Text'.

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Insulated Sectional Overhead Doors.

B. Electric Operators and Controls.

C. Operating Hardware, tracks, and support.

1.2 RELATED SECTIONS

A. Section 03300 - Cast-In-Place Concrete.

B. Section 04810 – Concrete Unit Masonry.

C. Section 05500 - Metal Fabrications.

D. Section 06114 – Wood Framing.

E. Section 07900 - Joint Sealants.

F. Section 08710 - Door Hardware.

G. Section 09900 - Paints and Coatings.

H. Section 11150 - Parking Control Equipment.

I. Section 16130 - Raceway and Boxes.

J. Section 16150 - Common Work Results for Electrical.

1.3 REFERENCES

A. ANSI/DASMA 102 - American National Standard Specifications for Sectional Overhead Type Doors.

1.4 DESIGN / PERFORMANCE REQUIREMENTS

A. Wiring Connections: Requirements for electrical characteristics.
   1. 115 volts, single phase, 60 Hz.
   2. 230 volts, single phase, 60 Hz.
   3. 230 volts, three phase, 60 Hz.
   4. 460 volts, three phase, 60 Hz.

B. Single-Source Responsibility: Provide doors, tracks, motors, and accessories from one manufacturer for each type of door. Provide secondary components from source acceptable to manufacturer of primary components.

1.5 SUBMITTALS

A. Submit under provisions of Section 01300.

B. Product Data: Manufacturer's data sheets on each product to be used, including:
   1. Preparation instructions and recommendations.
   2. Storage and handling requirements and recommendations.
   3. Installation methods.

C. Shop Drawings: Indicate plans and elevations including opening dimensions and required tolerances, connection details, anchorage spacing, hardware locations, and installation details.

D. Manufacturer's Certificates: Certify products meet or exceed specified requirements.

E. Operation and Maintenance Data.

1.6 QUALITY ASSURANCE

A. Manufacturer Qualifications: Company specializing in manufacturing products specified in this section with minimum five years documented experience.

B. Installer Qualifications: Authorized representative of the manufacturer with minimum five years documented experience.

C. Products Requiring Electrical Connection: Listed and classified by Underwriters Laboratories, Inc. acceptable to authority having jurisdiction as suitable for purpose specified.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Store products in manufacturer's unopened labeled packaging until ready for installation.

B. Protect materials from exposure to moisture until ready for installation.

C. Store materials in a dry, ventilated weathertight location.

1.8 PROJECT CONDITIONS

A. Pre-Installation Conference: Convene a pre-installation conference just prior to commencement of field operations, to establish procedures to maintain optimum working conditions and to coordinate this work with related and adjacent work.

1.9 WARRANTY

A. Warranty: Manufacturer's limited door warranty for 10 year against delamination of polyurethane foam from steel face and all other components for 1 year.

B. Warranty: Manufacturer's limited door and operators System warranty for 10 year against delamination of polyurethane foam from steel face and all other components for 3 years or 20,000 cycles, whichever comes first.

PART 2 PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturer: Overhead Door Corporation, 2501 S. State Hwy. 121, Suite 200, Lewisville, TX 75067. ASD. Tel. Toll Free: (800) 275-3290. Phone: (469) 549-7100. Fax: (972) 906-1499. Web Site: www.overheaddoor.com. E-mail: info@overheaddoor.com.

B. Substitutions: Not permitted.

C. Requests for substitutions will be considered in accordance with provisions of Section 01600.

2.2 INSULATED SECTIONAL OVERHEAD DOORS

A. Insulated Steel Sectional Overhead Doors: Model 591 Thermacore Insulated Steel Doors by Overhead Door™ Brand. Units shall have the following characteristics:

1. Door Assembly: Metal/foam/metal sandwich panel construction, with PVC thermal break and weather-tight ship-lap design meeting joints.
   a. Panel Thickness: 1-5/8 inches (41 mm).
   b. Exterior Surface: Ribbed, textured.
   c. Exterior Steel: .015 inch (.38 mm), hot-dipped galvanized.
   d. End Stiles: 16 gauge.
   e. Spring Counterbalance: Sized to weight of the door, with a helically wound, oil tempered torsion spring mounted on a steel shaft; cable drum of diecast aluminum with high strength galvanized aircraft cable. Sized with a minimum 7 to 1 safety factor.
      1) Standard cycle spring: 10,000 cycles.
      2) High cycle spring: 25,000 cycles.
      3) High cycle spring: 50,000 cycles.
      4) High cycle spring: 75,000 cycles.
      5) High cycle spring: 100,000 cycles.
   f. Insulation: CFC-free and HCFC-free polyurethane, fully encapsulated.
   g. Thermal Values: Tested installed assembly U-factor of 0.13 Btu/hr/SF degrees F; calculated section R-value of 14.86.
   h. Air Infiltration: 0.08 cfm at 15 mph; 0.08 cfm at 25 mph.
   i. Pass-Door: Provide with optional pass door.
   j. High-Usage Package: Provide with optional high-usage package.
   k. Partial Glazing of Steel Panels:
      1) 1/8 inch (3 mm) Acrylic glazing.
      2) 1/4 inch (6 mm) Acrylic glazing.
      3) 1/8 inch (3 mm) Clear Lexan glazing.
      4) 1/4 inch (6 mm) Clear Lexan glazing.
      5) 1/2 inch (12.5 mm) Clear Lexan Insulated glazing.
      6) 1/8 inch (3 mm) Tempered glass.
      7) 1/4 inch (6 mm) Tempered glass.
      8) 1/2 inch (12.5 mm) Tempered Insulating Glass.

9) 1/4 inch (6 mm) Wire glass.
10) 1/8 inch (3 mm) Double Strength glass.
11) 1/2 inch (12.5 mm) Double Strength Insulating Glass.
12) 1/8 inch (3 mm) Low E glazing.
13) 1/4 inch (6 mm) Low E glazing.
14) 1/2 inch (12.5 mm) Low E Insulated glazing.
15) 1/8 inch (3 mm) Solar Bronze glazing.
16) 1/4 inch (6 mm) Solar Bronze glazing.
17) 1/2 inch (12.5 mm) Solar Bronze Insulated glazing.
18) 1/8 inch (3 mm) Obscure glazing.
19) 1/4 inch (6 mm) Obscure glazing.
20) 1/2 inch (12.5 mm) Obscure Insulated glazing.
21) 1/4 inch (6 mm) Twin-Wall Polycarbonate (clear, bronze, white).
22) 3/8 inch (9.5 mm) Twin-Wall Polycarbonate (clear, bronze, white).
23) 5/8 inch (15.87 mm) Triple-Wall Polycarbonate (clear, bronze, white).

l. Full Glazed Aluminum Sash Panels:
1) 1/8 inch (3 mm) Acrylic glazing.
2) 1/4 inch (6 mm) Acrylic glazing.
3) 1/8 inch (3 mm) Clear Lexan glazing.
4) 1/4 inch (6 mm) Clear Lexan glazing.
5) 1/2 inch (12.5 mm) Clear Lexan Insulated glazing.
6) 1/8 inch (3 mm) Tempered glass.
7) 1/4 inch (6 mm) Tempered glass.
8) 1/2 inch (12.5 mm) Tempered Insulating Glass.
9) 1/4 inch (6 mm) Wire glass.
10) 1/8 inch (3 mm) Double Strength glass.
11) 1/2 inch (12.5 mm) Double Strength Insulating Glass.
12) 1/8 inch (3 mm) Low E glazing.
13) 1/4 inch (6 mm) Low E glazing.
14) 1/2 inch (12.5 mm) Low E Insulated glazing.
15) 1/8 inch (3 mm) Solar Bronze glazing.
16) 1/4 inch (6 mm) Solar Bronze glazing.
17) 1/2 inch (12.5 mm) Solar Bronze Insulated glazing.
18) 1/8 inch (3 mm) Obscure glazing.
19) 1/4 inch (6 mm) Obscure glazing.
20) 1/2 inch (12.5 mm) Obscure Insulated glazing.
21) 1/4 inch (6 mm) Twin-Wall Polycarbonate (clear, bronze, white).
22) 3/8 inch (9.5 mm) Twin-Wall Polycarbonate (clear, bronze, white).
23) 5/8 inch (15.87 mm) Triple-Wall Polycarbonate (clear, bronze, white).

2. Finish and Color:
a. Two coat baked-on polyester:
1) Interior color, white.
2) Exterior color, white.
3) Exterior color, brown.
4) Exterior color, tan.
5) Exterior color, gray.

b. Baked-on Trinar polyvinylidene fluoride high performance coating:
1) Exterior color, white.
2) Exterior color, brown.
3) Exterior color, beige.

3. Wind Load Design: Design as calculated in accordance with applicable code as follows:

a. Design pressure of _____ lb/sq ft (_____kPa).

4. Hardware: Galvanized steel hinges and fixtures. Ball bearing rollers with hardened steel races.
5. Lock:
   a. Interior mounted slide lock.
   b. Interior mounted slide lock with interlock switch for automatic operator.
   c. Keyed lock.
   d. Keyed lock with interlock switch for automatic operator.
   e. Locking mechanism designed to maintain security for exterior while permitting break out when impacted from the inside.
6. Weatherstripping:
   a. EPDM bulb-type strip at bottom section.
   b. Flexible Jamb seals.
   c. Flexible Header seal.
7. Track: Provide track as recommended by manufacturer to suit loading required and clearances available.
   a. Size:
      1) 2 inch (51 mm).
      2) 3 inch (76 mm).
   b. Type:
      1) Standard lift.
      2) Vertical lift.
      3) High lift.
      4) Low headroom.
      5) Follow roof slope.
8. Manual Operation: Pull rope.
9. Manual Operation: Chain hoist.
10. Electric Motor Operation: Provide UL listed electric operator, size and type as recommended by manufacturer to move door in either direction at not less than 2/3 foot nor more than 1 foot per second. Operator shall meet UL325/2010 requirements for continuous monitoring of safety devices.
    a. Entrapment Protection: Required for momentary contact, includes radio control operation.
       1) Pneumatic sensing edge up to 18 feet (5.5 m) wide. Constant contact only complying with UL 325/2010.
       2) Electric sensing edge monitored to meet UL 325/2010.
       3) Photoelectric sensors monitored to meet UL 325/2010.
    b. Operator Controls:
       1) Push-button operated control stations with open, close, and stop buttons.
       2) Key operated control stations with open, close, and stop buttons.
       3) Push-button and key operated control stations with open, close, and stop buttons.
       4) Flush mounting.
       5) Surface mounting.
       6) Interior location.
       7) Exterior location.
       8) Both interior and exterior location.
    c. Special Operation:
       1) Pull switch.
       2) Vehicle detector operation.
       3) Radio control operation.
       4) Card reader control.
       5) Photocell operation.
       6) Door timer operation.
       7) Commercial light package.

8) Explosion and dust ignition proof control wiring.

PART 3 EXECUTION

3.1 EXAMINATION

A. Do not begin installation until openings have been properly prepared.

B. Verify wall openings are ready to receive work and opening dimensions and tolerances are within specified limits.

C. Verify electric power is available and of correct characteristics.

D. If preparation is the responsibility of another installer, notify Architect of unsatisfactory preparation before proceeding.

3.2 PREPARATION

A. Clean adjacent surfaces thoroughly prior to installation.

B. Prepare surfaces using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions.

3.3 INSTALLATION

A. Install overhead doors and track in accordance with approved shop drawings and the manufacturer's printed instructions.

B. Coordinate installation with adjacent work to ensure proper clearances and allow for maintenance.

C. Anchor assembly to wall construction and building framing without distortion or stress.

D. Securely brace door tracks suspended from structure. Secure tracks to structural members only.

E. Fit and align door assembly including hardware.

F. Coordinate installation of electrical service. Complete power and control wiring from disconnect to unit components.

3.4 CLEANING AND ADJUSTING

A. Adjust door assembly to smooth operation and in full contact with weatherstripping.

B. Clean doors, frames, glass, and polycarbonate according to manufacturer’s instructions.

C. Remove temporary labels and visible markings. Do not remove polycarbonate care and maintenance label required to maintain warranty.

3.5 PROTECTION

A. Do not permit construction traffic through overhead door openings after adjustment and cleaning.

B. Protect installed products until completion of project.

C. Touch-up, damaged coatings and finishes and repair minor damage before Substantial Completion.

END OF SECTION